

Task 2 — BookStore Console Application (C# OOP and Advanced)

User Stories

As the owner of the bookstore, I want:

- A small command-line application that lets me manage my books, my customers, and the purchases they make.
- The application to reject any invalid input and tell me clearly what went wrong.
- The application to be easy to extend later without breaking what already works.

Requirements

- The application must let the user add, remove, search, and list books.
- The application must let the user register new customers and record purchases that contain more than one book.
- The application must validate every piece of user input and never crash from a bad value.
- The application must be designed to support new book formats in the future (for example, paperback, ebook, audiobook) without rewriting the existing code.
- The application must keep its data consistent at all times. Selling a book that is out of stock should not be allowed.
- The application must give the user clear feedback after every action.

Tasks

1. Design the domain model for the bookstore. Decide the classes, their relationships, and which OOP features you will use.
2. Build a working console menu that lets the user add, remove, search, and list books.
3. Support registering new customers and recording purchases that contain more than one book.
4. Validate all user input. Show clear, friendly error messages and never let the application crash.
5. Calculate the total revenue, the best-selling book, and the top customer.
6. Filter books by category, by author, and by price range.
7. Adding a new book format in the future must not require changing any code that already works.
8. Build a generic in-memory repository that can store and retrieve any entity in the system.

9. Allow applying any custom rule a developer chooses to a list of books — for example, applying a discount or a price adjustment.
10. Notify other parts of the system whenever a book runs out of stock.
11. Add helper methods to existing .NET types where they make your code cleaner.
12. **(Bonus)** Save all data to a file and reload it on startup. Use async file operations.
13. **(Bonus)** Allow multiple purchases to happen at the same time without breaking the stock count.

Submission Rules

- A working .NET 8 console application in a public GitHub repo.
- README must explain how to run the application and include sample screenshots of the menu in action.